

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |       |                      |                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|----------------------|------------------------|
| DS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Session | Queue | Question Information | Level 1   Challenge 52 |
| <p><b>Question description</b></p> <p>There is a bit string consisting of <math>n</math> bits. Then, there are some changes that invert one given bit. Your task is to report, after each change, the length of the longest substring whose each bit is the same.</p> <p><b>Constraints</b></p> <ul style="list-style-type: none"><li>• <math>1 \leq n \leq 2 \cdot 10^5</math></li><li>• <math>1 \leq m \leq 2 \cdot 10^5</math></li><li>• <math>1 \leq x_i \leq n</math></li></ul> <p><b>Input Format</b></p> <p>The first input line has a bit string consisting of <math>n</math> bits. The bits are numbered <math>1, 2, \dots, n</math>.<br/>The next line contains an integer <math>m</math>: the number of changes.<br/>The last line contains <math>m</math> integers <math>x_1, x_2, \dots, x_m</math> describing the changes.</p> <p><b>Output Format</b></p> <p>After each change, print the length of the longest substring whose each bit is the same.</p> |         |       |                      |                        |

✓ Logical Test Cases

Test Case 1

INPUT (STDIN)

001011  
3  
3 2 5

EXPECTED OUTPUT

4 2 3

Test Case 2

INPUT (STDIN)

101010  
4  
1 3 2 6

EXPECTED OUTPUT

2 4 2 2

✓ Mandatory Test Cases

Test Case 1

KEYWORD

void pull(int k,int  
l,int r)

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

550

Test Case 3

NLOC

59